Technische Hochschule Nürnberg

Fakultät Informatik

RaspberryPi CyberSec Lab: Development of a Penetration Testing Platform

Bachelorarbeit im Studiengang Elektro- und Informationstechnik

vorgelegt von

Jonas Schmitt

Matrikelnummer 3619551

Erstgutachter: Prof. Dr. Aykan D. Inan

Zweitgutachter: Jörg Arndt

© 2025

Dieses Werk einschließlich seiner Teile ist **urheberrechtlich geschützt**. Jede Verwertung außerhalb der engen Grenzen des Urheberrechtgesetzes ist ohne Zustimmung des Autors unzulässig und strafbar. Das gilt insbesondere für Vervielfältigungen, Übersetzungen, Mikroverfilmungen sowie die Einspeicherung und Verarbeitung in elektronischen Systemen.

Hinweis: Diese Erklärung ist mit Originalunterschrift (nicht gescannt) in das Papierexemplar der Abschlussarbeit fest einzubinden. Eine Spiralbindung ist nicht zulässig. Das digitale Exemplar enthält einen Scan der Erklärung mit Unterschrift.

Prüfungsrechtliche Erklärung der/des Studierenden

Angaben des bzw. der Studierenden:

Name: Vorname: Matrikel-Nr.:

Fakultät: Studiengang:

Semester:

Titel der Abschlussarbeit:

Ich versichere, dass ich die Arbeit selbständig verfasst, nicht anderweitig für Prüfungszwecke vorgelegt, alle benutzten Quellen und Hilfsmittel angegeben sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet habe. Außerdem versichere ich, dass der Hauptteil des Papierexemplares und des digitalen Exemplares identisch sind. Das digitale Exemplar enthält, falls gefordert, lediglich weitere Anlagen.

Ort, Datum, Unterschrift Studierende/Studierender

Erklärung der/des Studierenden zur Veröffentlichung der vorstehend bezeichneten Abschlussarbeit

Die Entscheidung über die vollständige oder auszugsweise Veröffentlichung der Abschlussarbeit liegt grundsätzlich erst einmal allein in der Zuständigkeit der/des studentischen Verfasserin/Verfassers. Nach dem Urheberrechtsgesetz (UrhG) erwirbt die Verfasserin/der Verfasser einer Abschlussarbeit mit Anfertigung ihrer/seiner Arbeit das alleinige Urheberrecht und grundsätzlich auch die hieraus resultierenden Nutzungsrechte wie z.B. Erstveröffentlichung (§ 12 UrhG), Verbreitung (§ 17 UrhG), Vervielfältigung (§ 16 UrhG), Online-Nutzung usw., also alle Rechte, die die nicht-kommerzielle oder kommerzielle Verwertung betreffen.

Die Hochschule und deren Beschäftigte werden Abschlussarbeiten oder Teile davon nicht ohne Zustimmung der/des studentischen Verfasserin/Verfassers veröffentlichen, insbesondere nicht öffentlich zugänglich in die Bibliothek der Hochschule einstellen.

Hiermit ☐ genehmige ich, wenn und soweit keine entgegenstehenden Vereinbarungen mit Dritten getroffen worden sind,

☐ genehmige ich **nicht,**

dass die oben genannte Abschlussarbeit durch die Technische Hochschule Nürnberg Georg Simon Ohm, ggf. nach Ablauf einer mittels eines auf der Abschlussarbeit aufgebrachten Sperrvermerks kenntlich gemachten Sperrfrist

von Jahren (0 - 5 Jahren ab Datum der Abgabe der Arbeit),

der Öffentlichkeit zugänglich gemacht wird. Im Falle der Genehmigung erfolgt diese unwiderruflich; hierzu wird der Abschlussarbeit ein Exemplar im digitalisierten PDF-Format an die Betreuer übermittelt. Bestimmungen der jeweils geltenden Studien- und Prüfungsordnung über Art und Umfang der im Rahmen der Arbeit abzugebenden Exemplare und Materialien werden hierdurch nicht berührt.

Ort, Datum, Unterschrift Studierende/Studierender

Datenschutz: Die Antragstellung ist regelmäßig mit der Speicherung und Verarbeitung der von Ihnen mitgeteilten Daten durch die Technische Hochschule Nürnberg Georg Simon Ohm verbunden. Weitere Informationen zum Umgang der Technischen Hochschule Nürnberg mit Ihren personenbezogenen Daten sind unter nachfolgendem Link abrufbar: https://www.th-nuernberg.de/datenschutz/

Kurzdarstellung

Diese Arbeit befasst sich mit der Entwicklung einer Testplatform für Cyberangriffe mit dem Fokus auf Funkverbindungen. Im Rahmen den Projekts wurde auf Basis von RaspberryPi OS Software entwickelt, um Umgebungen zu schaffen, in denen verschiedene Cyberangriffe getestet werden können.

Abstract

Contents

1. Introduction **1**
1.1. This is an Important Section 1
1.1.1. And an even more important subsection 1

2. Basics **3**
2.1. Cybersecurity 3
2.2. WiFi History 3
2.3. MAC layer 4
2.3.1. Architecture 4
2.3.2. MAC Frames 4
2.3.3. Medium Access 6
2.3.4. Connection Establishment and Termination 6
2.3.5. PHY layer 6
2.3.6. Security 6

3. Architecture **7**
3.1. Hardware 7
3.2. Software 8

4. Worklog **11**
4.1. Abc 11

5. Wifi Attacks **13**
5.1. Abc 13

6. Penetration Testing **15**
6.1. Abc 15

7. Educational Use **17**
7.1. abc 17

8. Outlook **19**

9. Summary **21**

A. Supplemental Information **23**

List of Figures 25

List of Tables 27

List of Listings 29

Bibliography 31

Glossary 33

Chapter 1.

Introduction

You may have read about similar things in [**?**]. You can also write footnotes.[1]

1.1. This is an Important Section

It is possible to reference glossary entries as library as an example.

1.1.1. And an even more important subsection

[1] Footnotes will be positioned automatically.

Chapter 2.

Basics

2.1. Cybersecurity

Through the progression of digitalization, Cybersecurity is of ever increasing relevance to the security of IT-Systems. An increasing number of devices are connected to the internet, making them part of the cyberspace and therefore potentially vulnerable to cyberattacks. Almost all sectors of our lives are at risk, when our homes, infrastructure, media devices, industries and governments are connected to the internet.

Cybersecurity aims to protect these devices by achieving the well-known CIA triad, which is an acronym for Confidentiality, Integrity, and Availability. [Oriy 17, chapter Attacking Integrity Controls]

2.2. WiFi History

In 1985 the american Federal Communications Commission ruled serveral bands in the 2.4 GHz radio band to be unlicensed, meaning everyone can broadcast radiosignals in these frequencies. At the beginning of the 90's, this lead to the development of different wireless communication protocols by multiple companies. To ensure interoperability between these protocols, the IEEE 802.11 standard was created and released, and WLAN was born. In 1999 the Wi-Fi Alliance (WFA) was founded to further improve compatibility, nowadays consisting of over 750 international companies. [Sank 21, page 1-3], [Wiki 25]

Since its inception over 25 years ago, Wi-Fi has gone through constant development, leading to six major generations of the technology. Improvments in speed, efficiency and security have lead to the technology becoming a "fundamental utility [...], that everyone [...] expects to be available everywhere" [Sank 21, page 1]. WiFi connects billions of devices, like computers, smartphones, game consoles or IoT devices to the internet, generating a value of around 4.9 trillion USD worldwide [Sank 21, page 1].

WiFi operates at the physical (PHY) and the medium access control (MAC) layer of the OSI model, sending ethernet frames over radio waves in the 2.4 and 5 Ghz frequency band. The Mac layer is responsible for seting rules on tranmission and reception of data, connection management and power management. Data sent on the MAC layer then gets converted to radio waveforms by the PHY layer, which is also responsible for converting received data back to the MAC protocol. [Sank 21, page 4-7]

2.3. MAC layer

2.3.1. Architecture

Commonly WiFi networks have access points (AP), which are connected to the internet via ethernet, and stations (STA), which are provided with internet access over WiFi by the AP. WiFi has a MAC layer, which is designed to be flexible and can be configured in the following topologies:

- **BSS**: Basic Service Set
- **IBSS**: ad hoc / peer-to-peer / Independent BSS
- **WDS**: Wireless Distribution System
- **MBSS**: Mesh BSS

BSS describes the most common and basic configuration, where multiple STAs are connected to one AP, which has wired connection to the internet. IBSS networks dont have any APs, instead the STAs are connected directly to each other. WDS can be used, when the reach of an AP needs to be extended, but there is no wired connection available. A secondary AP can be installed which receives backhaul connection from another AP. Mesh Networks are complex BSS networks with multiple APs, which the STAs can hop between. This type of network can improve the signal coverage and resilience and is often used in companies or public institutions with many STAs. [Sank 21, page 8-9]

2.3.2. MAC Frames

When data is sent over the MAC layer, it gets divided into chunks, which are called MAC Frames. MAC Frames have three components: the header, the body and the frame check sequence. The header contains information about the source and destination, the sequence and serveral control functions. In a BSS network it has the following functions:

- **Frame Control** contains 16 Bits which serve different control functions

  protocol version describes which version of the MAC protocol this frame uses

  type describes the type of MAC frame is beeing used: there are data, management, control and acknowledgment frames

  subtype describes the subtype of the frame, e.g. association request or beacon frame

  from/to ds indicates the direction of travel

  more fragment indicates more following fragments

  retry indicates that the current frame is a retransmission

  power management gives information about the power state of the STA

  more data indicates more frames are to be transmitted

  protected frame indicates whether a frame is encrypted

  order set to 1 if the order of processing is important

- **duration** indicates the time of the transmission, therefore how long the medium is occupied

- **adress 1** contains the receiver address (RA), which can be the MAC address of the destination STA or the BSSID of the AP, depending on the direction, the frame travels

- **adress 2** contains the transmitter adress (TA), which can be the BSSID or the source address.

- **adress 3** contains either the source or destination address

- **sequence control**

  fragment number ensures the correct processing sequence of the fragments

  sequence number number of the frame

The Frame body then contains the payload, meaning the actual information to be sent. Every type of MAC frame is configured a different way to suit their specific function. "While data frames are meant to carry data, management frames are used for connection management and control frames [...] for triggering specific actions. Control frames have a smaller MAC header and no payload data [...]. Management frames have a structured payload composed of some fixed length fields and one or more variable length fields called information elements (IEs)" [Sank 21, page 12]

After the frame body comes the frame control sequence (FCS), which ensures the integrity of the frame. The FCS is a cyclic redundancy check (CRC), which is number calculated from the frame header and body content. When the frame is compromised in traffic, the result of the CRC changes, therefore the error can be identified. [Geek 23] [Sank 21, page 10-12]

2.3.3. Medium Access

In LAN and WLAN networks all devices share the same medium, therefore only one device at a time can transmit information. Ethernet networks use full-duplex transceivers, meaning a device can transmit and receive data at the same time. This allows the transceiver to transmit and simlutaniously listen for collisions with other frames. Collision detection is quite effective, but can not be used in WiFi networks, because WiFi transceivers are only capable of half-duplex tranmission. WiFi instead uses a medium access control referred to as carrier sense multiple access collision avoidance (CSMA-CA).

When using this technique, the medium gets blocked during transmission and is freed by an acknowledgment frame, sent by the receiving device. All STAs start off in the random backoff state, where they listen for traffic. After the random backoff counter has expired and no traffic is present, the STA will start transmission. CSMA-CA can be described by the way "humans converse in a group, where every individual would first listen to check if someone else is talking, wait for the person talking to finish, and then start talking after a brief random pause" [Sank 21, page 12]. [Oriy 17, chapter Attacking Integrity Controls] [Sank 21, page 12-17]

2.3.4. Connection Establishment and Termination

2.3.5. PHY layer

The first generations of WiFi used transmit in the 2.4 Ghz band, but due to it beeing used by many other devices, such as cordless phones, computer peripherals, bluetooth devices or microwave ovens, WiFi 4 started using the 5 Ghz band, in order to improve connectivity.

2.3.6. Security

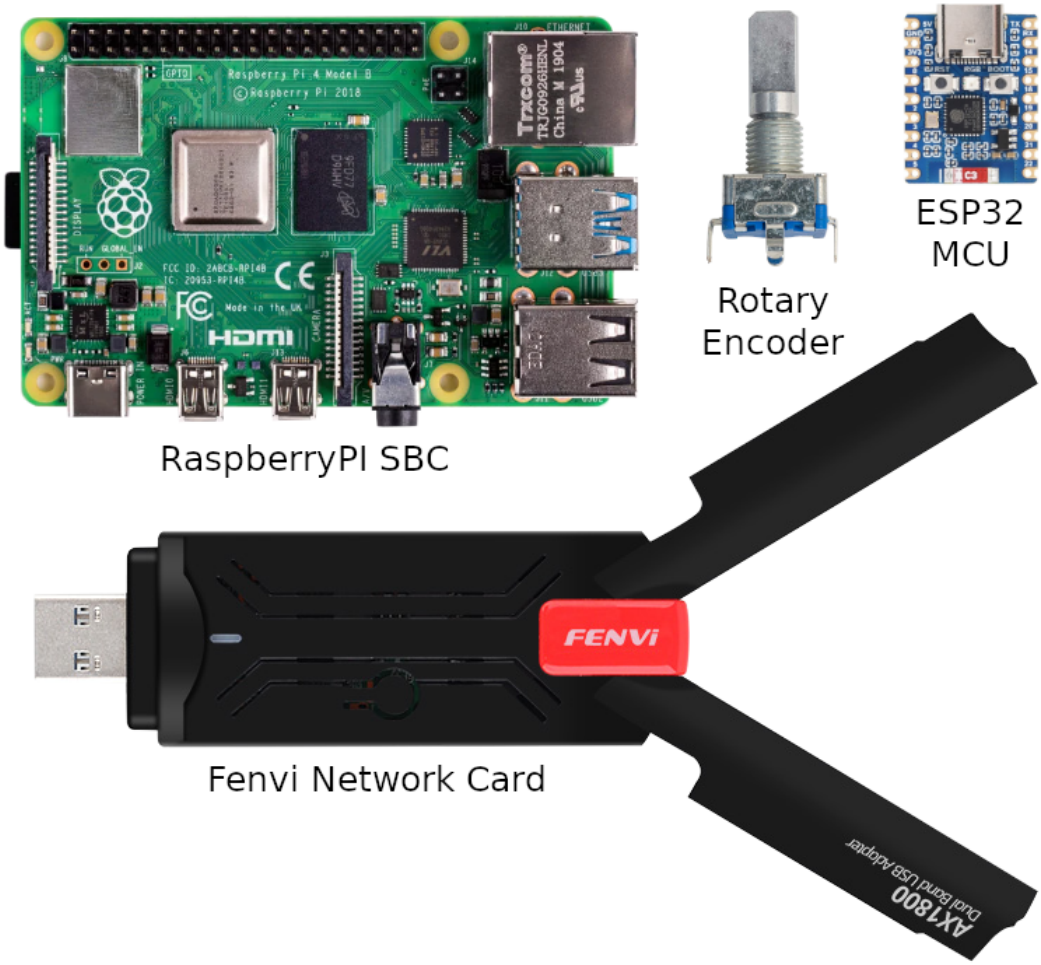
Chapter 3.

Architecture

3.1. Hardware

The Project is realized using a RaspberryPI as the central device, connected to which are an external network card, an ESP32 microcontroller, a display and a rotary encoder for user input.

As the main computer the RaspberryPI 4B with 8GB of RAM was chosen due to its ease of use, linux based operating system and widely available documentation.



3.2. Software

The software of the project is mostly modular, it consists of a central program, which runs the user interface and then executes various shell scripts, with which the functions are implemented. When executing the setup script, the main program, which is written in C++, gets compiled and the executable gets called by the start.sh script. The program then displays the user interface, which is a simple interface, printed into the console. Each page of the UI is a "Menu" object, each menu has a title, options of the type "MenuOptions", a variable to track the currently selected option, a method to add options to a menu and methods for navigation, input capture and displaying. MenuOption objects have a name, as well as an action of the std::function type, which can be executed by calling the execute() method. In the main of the program, the constructors for menues and the addOption() methods are beeing used, to create the different menues and their submenues. The program starts by call-

ing the navigate() function of the main menu, which then displays all available submenues and menu options. All submenues work in the same way as the main menu, getting their respective navigate() called upon selection. During navigation the UI peridiodically prints the available options, using unicode to highlight the currently selected option as well as to give contrast to the title of the menu.

For input the device uses a rotary encoder with a pushbutton, turning the encoder moves the selection up or down in the menues, the pushbutton executes the selected option. The encoder is connected to the RaspberryPIs GPIO interface and the input signals are read and processed by the encoder program. The start.sh script starts both programs and pipes the stdout of encoder into the stdin of main. When an action of the encoder is detected, the according command gets sent to main, the commands beeing the unicodes for the up/down arrows and the enter key on a keyboard. For example, the User turns the the encoder one click clockwise, therefore main receives one downwards arrow symbol and the selections moves to the next option below. Because no special commands are utilized for navigation, the program can be easily tested on any computer by navigating with the keyboard.

When an action is executed, main calls the according shell script with the system() function. The wifiActivate.sh script is used to open wifi hotspots, where the user can choose the type of Wifi standard beeing used. It receives an input command and then establishes a WEP, WPA, WPA2 or WPA3 network.

Listing 3.1: extract of wifiActivate.sh

```
#!/bin/bash

scriptdir="$(dirname␣"$0")"

read -n 4 connection

if [ "$connection" == "WEP" ]; then
   exists=$(nmcli connection show | sed -n '/Wifi-WEP␣/p')
   if [ -z "$exists" ]; then
      sudo nmcli device wifi hotspot con-name "Wifi-WEP" ssid "WEPtest" password
```

With the wifiReset.sh script, the user can deactivate the active network. Once a network has been cracked, the wifiNewPassword.sh script can be used to give a new random password to the selected network. The script the pulls a random password out of a password list, which contains popular passwords that have been filtered to work with the specification of each network standard. To display the current status of the network, meaning its security standard, SSID and password, the wifiStatus.sh script is used.

```
#include <string.h>
int x = 1;
```

```
if(x == 1){

    printf("x is 1.");
}
```

Listing 3.2: This is an example of inline listing

You can also include listings from a file directly:

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Listing 3.3: This is an example of included listing

Chapter 4.

Worklog

4.1. Abc

Chapter 5.

Wifi Attacks

One of the earliest and most commonly used forms of attack used to target access control is a technique known as war driving. This technique, simply put, is to use a wireless enabled device along with specialized software used to detect or probe wireless networks that have come with in range of the wireless device. What makes war driving so popular and so effective is the fact that many computer users, both personal and business, have been deploying 802.11 wireless access points for years with little regard for security. The deployment of a wireless network that allows the user to roam has taken precedence over taking measures to secure their access points and devices against potential attack. In response to this deployment of wireless networks that are misconfigured or insecure, we have a category of attackers that engage in this practice known as war driving. Those who engage in this activity will build custom rigs built out of a combination of hardware and software, with the intent of cruising public venues to find both wireless access points and devices in some cases. These individuals may just be interested in locating and targeting a specific network for their own ends or, in some cases, they may even share this information to an online database or website where any visitor can search for these access points.

[Oriy 17]

5.1. Abc

Chapter 6.

Penetration Testing

6.1. Abc

Chapter 7.

Educational Use

7.1. abc

Chapter 8.

Outlook

Chapter 9.

Summary

Appendix A.

Supplemental Information

Anleitung zum Raspi-CyberSec-Lab

Jonas Schmitt

6. März 2025

Inhaltsverzeichnis

1 Allgemeine Informationen **2**

2 Voraussetzungen **2**

3 Installation **2**

4 Benutzeranleitung **3**

4.1 Übersicht Hauptmenü 3

4.2 Navigation 3

4.3 Menü Übersicht 3

List of Figures

List of Tables

List of Listings

3.1. extract of wifiActivate.sh 9
3.2. This is an example of inline listing 9
3.3. This is an example of included listing 10

Bibliography

[Geek 23] GeeksforGeeks. “IEEE 802.11 mac Frame”. Mar 2023.

[Oriy 17] S.-P. Oriyano. *Kali Linux Wireless Penetration Testing Cookbook: Identify and assess vulnerabilities present in your wireless network, Wi-Fi, and bluetooth enabled devices to improve your wireless security.* Packt Publishing, 2017.

[Sank 21] S. G. Sankaran and S. R. Gulasekaran. *Wi-Fi 6: Protocol and Network.* Artech House, 2021.

[Wiki 25] Wikipedia. “Wi-Fi”. Mar 2025.

Glossary

library A suite of reusable code inside of a programming language for software development. i, 1

shell Terminal of a Linux/Unix system for entering commands. i